

Research Statement

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My research develops statistical methodology for complex, heterogeneous, and high-dimensional data arising in genomics, mobile health, and biomedical sciences. A central theme is integrating information from multiple data sources under constraints such as missingness, high dimensionality, and computational scalability. My contributions span four interconnected areas: *multi-source data integration with missing data*, *statistical genetics and genomics*, *mobile health and wearable device data*, and *causal mediation analysis*. My work has appeared in leading journals including the *Annals of Statistics*, *Journal of the American Statistical Association* (JASA), *Biometrika*, and *Annals of Applied Statistics*, supported by grants from the National Science Foundation (NSF) and the National Institutes of Health (NIH).

Multi-Source Data Integration and Missing Data

With the proliferation of multisource and multimodality data, block-wise missingness, where entire measurement blocks are absent for a subset of subjects, has become a central analytical challenge. I developed a multiple block-wise imputation (MBI) approach that integrates imputations from both complete and incomplete observations by constructing estimating equations over all possible missing patterns and combining them via the generalized method of moments (Xue and Qu, 2021). The method outperforms existing approaches on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset in terms of prediction error. This work was published in *Journal of the American Statistical Association*. Building on the MBI framework, I also proposed a multiple-imputed Dantzig selector (MDS) for high-dimensional inference under a semi-supervised setting in which outcomes are observed for only part of the samples (Xue et al., 2025). A bias-corrected MDS estimator, derived through projected estimating equations, enjoys asymptotic normality and enables valid confidence interval construction.

More recently, I developed identifiable deep latent variable models for missing not at random (MNAR) data (Xie et al., 2024), under revision at the *Journal of Machine Learning Research*. Under a conditional no self-censoring assumption given latent variables, this work establishes nonparametric identifiability of the complete data distribution and introduces a new importance-weighted autoencoder algorithm for MNAR imputation and data generation. Experiments on HIV clinical data and recommendation system data demonstrate substantial advantages over state-of-the-art deep generative methods. I also co-authored a book chapter on generative models for missing data in *Applications of Generative AI* (Springer, 2024).

In collaborative work on multi-response regression, I developed an empirical Bayes data integration framework for coefficient estimation across multiple data sources (Chakraborty and Xue, 2026). Rather than imposing sparsity or low-rank structure, which are difficult to verify in practice, I derived an asymptotically optimal linear shrinkage estimator by minimizing relative savings loss for covariance estimation. The proposed estimator works well under both sparse and dense, low-rank and non-low-rank parameter settings. The method demonstrates superior prediction performance in tissue-wide association studies (TWAS).

Statistical Genetics and Genomics

A major focus of my research is multi-tissue expression quantitative trait loci (eQTL) analysis using the Genotype-Tissue Expression (GTEx) dataset, which contains RNA-seq data from over 45 tissues of more than 800 donors. I developed an empirical Bayes regression model for predicting the effects of single nucleotide polymorphisms (SNPs) on gene expression (Xue and Li, 2026), employing a mixture prior that combines a multivariate Gaussian distribution with a point mass at zero to accommodate tissue-specific genetic effects. The method asymptotically achieves minimum Bayes risk, outperforms competing multi-tissue approaches in gene expression prediction on held-out GTEx data, and reveals that while genetic effects are broadly shared across tissues, effect sizes could vary substantially.

I have also established theoretical foundations for linkage disequilibrium score regression (LDSC), a widely used tool for genome-wide association study (GWAS) summary statistics analysis. Working within a fixed-effect data integration framework, I accounted for genome-wide dependence in high-dimensional GWAS summary statistics and block-diagonal structure in estimated LD scores, clarifying when asymptotic normality of LDSC-based estimators is achievable and extending LDSC to cross-ancestry analyses by modeling LD pattern disparities across global populations (Xue and Zhao, 2025). Theoretical findings are validated on UK Biobank data, and this work appeared in the *Annals of Statistics*. I further contributed to the BIGA cloud computing platform for bivariate cross-trait genetic architecture analysis, accepted at *Journal of the American Statistical Association* (Li et al., 2026), and to PennPRS, a centralized platform for polygenic risk score training in precision medicine.

Beyond GWAS, I have investigated network-based imaging genetics approaches that integrate brain connectivity data with genomic information to identify genetic variants associated with neuroimaging phenotypes.

Mobile Health and Wearable Device Data

Mobile health data present significant analytical challenges: they are heterogeneous, irregularly sampled, multi-resolution, and longitudinal. I proposed an individualized dynamic latent factor model for integrating multiple irregular multimodality time series, which interpolates unsampled measurements by mapping multi-resolution observations to a shared latent space and capturing heterogeneous longitudinal dynamics through individualized latent factors (Zhang et al., 2024). Published in *Biometrika*, this work provides theoretical bounds of the proposed method on integrated interpolation error and B-spline approximation convergence rates, and demonstrates superior performance over existing methods on smartwatch data.

I have extended this research through proposing a generalized heterogeneous functional model for large-scale mobile health data (Sun et al., 2024), under revision at *JASA*. We leveraged a pairwise fusion penalty to identify latent subgroups with distinct time-varying effects of physical activity on disease outcomes. Applied to over 79,000 UK Biobank participants, the proposed method uncovers subgroup-specific activity timing effects on mental disorders and Parkinson’s disease risk. A related project addresses early Parkinson’s disease detection from wrist-worn accelerometry (Xie et al., 2025), where we leveraged pretrained sleep staging

and activity recognition models to extract behavioral states, adopted hierarchical nonstationary Markov chains to model circadian transition dynamics, and used gradient-boosted Cox regression to estimate time-to-diagnosis in the UK Biobank study.

Causal Mediation Analysis

A major theme of my research is developing methods for understanding complex causal mechanisms in high-dimensional biomedical studies. I proposed a heterogeneous mediation analysis approach that allows mediation effects to vary across latent subpopulations (Xue et al., 2022), going beyond standard homogeneous frameworks. Applied to a predominantly African American cohort, this method uncovered heterogeneous epigenomic pathways linking traumatic stress, DNA methylation, and post-traumatic stress disorder (PTSD) risk, published in *JASA*. I have extended this methodology to longitudinal settings through a time-varying mediation analysis framework for incomplete data, combining structural equation modeling with multiple blockwise imputation and generalized method of moments estimation to handle high-dimensional repeated measurements and non-monotone missingness (Wei et al., 2025), published in the *Annals of Applied Statistics*.

I have also contributed to causal inference for omics studies, developing joint estimators for treatment effects on multiple correlated omics outcomes with false discovery rate (FDR) control that accounts for outcome dependence in randomized studies (Deek et al., 2025). In a related application using the ADNI dataset, I characterized causal pathways from depressive symptom profiles to Alzheimer’s disease risk, finding that memory-related symptoms affect cognitive decline through hippocampal volume and cerebral glucose metabolism (Zhang et al., 2025).

Summary and Future Directions

My research has advanced statistical methodology across data integration, high-dimensional inference, statistical genetics, mobile health analytics, and causal mediation analysis, with publications in top journals. Looking ahead, I plan to advance my work in three directions.

First, I will develop federated and transfer learning frameworks that account for privacy constraints and distributional heterogeneity across distributed health data studies. This includes methods for handling feature mismatch when variable sets differ across sites, and theoretical guarantees on estimation error under heterogeneous local distributions. The motivating applications are multi-site wearable studies and electronic health records in different hospitals, where patient-level data cannot be centralized.

Second, I am investigating large language models (LLM) as statistical proof assistants, examining their capabilities and limitations on research-level proof tasks spanning high-dimensional genomic inference, dependence modeling, transfer learning, and differential privacy. The goal is to identify practical workflows and shared proof-strategy repositories that effectively integrate AI tools into statistical research and graduate education.

Third, I will develop methods for synthetic data generation and multi-modal integration of imaging, genomic, clinical, and wearable data, enabling privacy-preserving analysis and complementary information fusion for disease risk prediction.

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